

Vibe coding

Internals Building your own vibe-coding tool from scratch

Vibe coding != AI Assistant

- Traditional coding + AI help you write/own the architecture, review diffs, and use AI as an assistant.
- Vibe-coding you lean heavily on the model to produce most of the implementation, and you steer via prompts + quick feedback loops (run/test/fix) rather than line-by-line scrutiny.



Focus & Goal

- No debate whether vibe coding is “good” or “bad”.
- Take an LM as a black box, no LLM internals, no math.
- Know how to build a coding agent from scratch using LLM

It's a good guideline / check list to build agents for your practical tasks.

Vibe Coding Tools

- Claude Code
- Cursor AI
- OpenAI Codex
- ...
- Gemini CLI
- Qwen CLI

Gemini CLI

- Open Source: <https://github.com/google-gemini/gemini-cli>
- gemini-2.5-pro / gemini-3-pro are good enough
- Hides nothing, prompt processing is done inside a client's application

System prompt

```
src/core/prompts.ts
```

```
"Analyse my application. Find all the security issues and fix them".length = 65 / 13  
turn.chat.systemInstruction.length = 16335 / 3518 (250 times larger)
```

You are an interactive CLI agent specializing in software engineering tasks.
Your primary goal is to help users safely and efficiently, adhering strictly to the following :
and utilizing your available tools.

Core Mandates

- ****Conventions:**** Rigorously adhere to existing project conventions when reading or modifying. Analyze surrounding code, tests, and configuration first.
- ****Libraries/Frameworks:**** NEVER assume a library/framework is available or appropriate. Verify its established usage within the project (check imports, configuration files like 'package.json', 'Cargo.toml', 'requirements.txt', 'build.gradle', etc., or observe neighboring files) before en...

Primary Workflows

Software Engineering Tasks

...

System prompt structure

- Setup, Core Mandates, AGENTS.md / GEMINI.md
- Available Sub-Agents (codebase_investigator) & Tools
- Primary Workflows
 - Software Engineering Tasks
 - New Applications
- Operational Guidelines
 - Shell tool output token efficiency
 - Tone and Style (CLI Interaction)
 - Security and Safety Rules
 - Tool Usage
 - Interaction Details
- Sandbox info
- Git Repository
- Final Reminder

Prompting

- **LLM & settings**
- General Prompt
- Prompt Techniques
 - Few-shot prompting [2020]
 - Chain Of Thoughts [2022]
 - RAG [2020]
 - Structured Output [2024]
 - ReAct [2023]
- Tools / MCP [MCP: late 2024]
- Agent Memory / Compression
- Misc

Prompting

- LLM & settings
- **General Prompt: set the task, be clear & explicit, focus on what should be done**
- Prompt Techniques
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Few-shot prompting

Is used in the final report in CodeInvestigator, prompt compressor, command output redirection.

STANDARD PROMPTING

Translate English to French:

cheese =>

FEW-SHOT PROMPTING

Translate English to French:

see otter => loutre de mer

peppermint => menthe poivrée

plush giraffe => girafe peluche

cheese =>

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Chain of Thoughts

Zero-shot CoT / Few-shot CoT

FEW-SHOT PROMPTING

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

A: The answer is 27. ✖

CHAIN OF THOUGHTS

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 balls each is 6 more tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more. So they have $3 + 6 = 9$. The answer is 9. ✔

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RAG: Retrieval-Augmented Generation

STANDARD PROMPTING

Q: What exactly is written on the sign at the gates of Moria?

A: A password is "Friend".

RAG PROMPT

Q: What exactly is written on the sign at the gates of Moria?
Use the genuine text from Context to answer.

Context:

‘The words are in the elven-tongue of the Elder Days,’ answered Gandalf. ‘But their importance to us. They say only: The Doors of Durin. Speak, friend, and enter. And underneath it, in smaller, fainter writing: “Celebrimbor of Hollin drew these signs”’

A: From the provided Context, Gandalf said, “The Doors of Durin, Lord of Moria. Speak, friend, and enter. And underneath it, in smaller, fainter writing: “Celebrimbor of Hollin drew these signs”’

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Structured Output

STANDARD PROMPT

Q: ... Your cards: Ah Qh Board (flop): Kh 7d 2h If you CALL, call amount is 6.

A: From A♥ Q♥ on K♥ 7♦ 2♥ (flop), there are 47 unseen cards and 1,081 possible turn+river combinations. Your final best 5-card hand by the river can be ... So, I decide to CALL (6)

STANDARD PROMPT + OUTPUT FORMAT

Q: ... Your cards: Ah Qh Board (flop): Kh 7d 2h If you CALL, call amount is 6.

```
class Action(BaseModel):  
    decision: Literal["FOLD", "CALL"]  
    amount: confloat(ge=6, le=100)  
    reason: str
```

```
Action.model_json_schema()
```

```
{  
    "decision": "CALL",  
    "amount": 6.0,  
    "reason": "From A♥ Q♥ on K♥ 7♦  
}
```

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LLM Query

There are libraries that proxy different APIs (i.e. litellm)

BEFORE: TEXT

```
Q: You are a smart Software Engineer.  
Check the project and make a list of top code issues ...
```

AFTER: JSON API (OPENAI FORMAT V1/V2)

```
{  
  "role": "system",  
  "content": "You are a smart Software Engineer.",  
},  
{  
  "role": "user",  
  "content": "Check the project and make a list of top code issues ..."  
}
```

```
Ok, there is a single place where SQL injection is possible
```

```
{  
  ...  
  "content": [  
    {  
      "type": "output_text",  
      "text": "Here is the patch",  
      "annotations": []  
    }  
  ]  
  ...  
}
```

Tool calls

Generated text is a list of JSON messages.

Message types are:

- reasoning
- text
- tool_call

```
{
  "type": "tool_call",
  "name": "delegate_to_agent",
  "arguments": {
    "agent_name": "codebase_investigator",
    "objective": "Check the project and make a list of top code issues ..."
  }
}
```

Gemini CLI Tools

Gemini CLI has 10 tools + CodeBase Investigator Sub-Agent

- `google_web_search`: Performs a web search using Google Search (via the Gemini API) results. This tool is useful for finding information on the internet based on a query.
- `web_fetch`: Processes content from URL(s), including local and private network addresses embedded in a prompt. Include up to 20 URLs and instructions (e.g., summarize, extract) directly in the 'prompt' parameter.
- `read_file` / `write_file`
- `glob` (aka FindFiles): Efficiently finds files matching specific glob patterns (e.g., `src/`) returning absolute paths sorted by modification time (newest first). Ideal for quickly listing their name or path structure, especially in large codebases.
- `run_shell_command`
- `delegate_to_agent`: Delegates a task to a specialized sub-agent ...

MCP

Anthropic introduced / open-sourced MCP on November 25, 2024. [Specification, 2025-11-25](#)

JSON-RPC based protocol

Some tools may be executed somewhere else, be accessible by multiple agents.

Examples:

- [AWS Documentation](#)
- [Trino MCP Server](#)
- [User Feedback](#)

MCP: Context 7

Has 2 methods:

- search: search library id by name + description
- context: get relevant documentation for the library by library id and user query

```
"mcpServers": {  
  "context7": {  
    "httpUrl": "https://mcp.context7.com/mcp",  
    "headers": {  
      "CONTEXT7_API_KEY": "YOUR_API_KEY",  
      "Accept": "application/json, text/event-stream"  
    }  
  }  
}
```

AGENTS.md

Always use context7 when you need code generation, you should automatically use the Context7 MCP tools to resolve library id and get library docs without me having to explicitly ask.

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ReAct

- Call LLM in a loop
- Response must be a JSON of a specific format [Structured Output].
- One of the attributes is `next_tool_name` .
- Run loop execution until `next_tool_name = finish` or `max_iter` reached.
- Message types are
 - reasoning
 - text
 - tool_call

The final prompt

The user instructions part of the last request.

The LLM reply ends the ReAct loop and reports the summary.

```
[
  {"role": "user", "parts": [
    {"type": "text", "content": "This is the Gemini CLI. We are setting up the context for our chat. Please provide the following information: // date, current directory, OS type, list of files (up to 200) in the current directory"},
    {"role": "user", "parts": [{"type": "text", "content": "Check the project and make a list of top code quality issues"}]}
  ],
  {"role": "system",
   "parts": [
    {"type": "text", "content": "I will delegate this task to the `codebase_investigator` ..."},
    {"type": "tool_call", "name": "delegate_to_agent", "arguments": {...}},
    {"role": "user", "parts": [
      {"type": "tool_call_response", "response": "<json output>"},
      {"role": "system", "parts": [{"type": "text", "content": "The analysis of your codebase is complete. Please provide a summary of the findings."}]}
    ]},
    {"role": "user", "parts": [{"type": "text", "content": "No, put all findings into an issues.md file"}]}
  ],
  {"role": "system", "parts": [
    {"type": "text", "content": "Of course. I will create an `issues.md` file summarizing the findings."},
    {"type": "tool_call", "name": "write_file", "arguments": "..."}
  ]},
  {"role": "user", "parts": [{"type": "tool_call_response", "response": '{"output": "Successfully created issues.md file with findings summary."}'}]}
]
```

ReAct

- Planning: implicit
- Tools:
 - basic 10 tools
 - code investigator sub-agent
 - mcp servers
- Memory:
 - message history

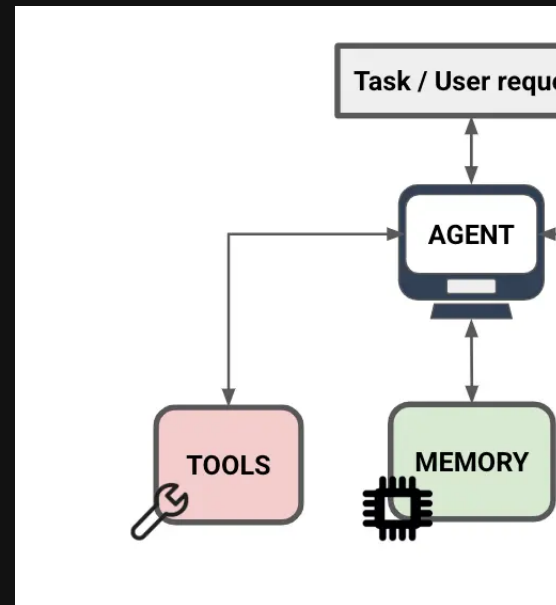
Agent

Implicit + explicit memory

```
ChatCompressionService
```

```
prompts.ts:getCompressionP
```

```
<state_snapshot> XML Objec
```



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Misc

- Sub-Agents: prompt grows like a snowball, sub-agents help to keep prompts smaller.
- Model Router: Gemini CLI picks between `gemini-2.5-pro` , `gemini-2.5-flash` , `gemini-2.5-flash-lite` , `gemini-2.5-pro-preview` is available. `gemini-2.5-flash-lite` is used for the router prompt.
- Skills: a new feature, introduced in 2026

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References

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- REACT : Synergizing Reasoning and Acting in Language Models <https://arxiv.org/pdf/>
- Chain-of-Thought Prompting Elicits Reasoning in Large Language Models <https://arxiv.org/pdf/2402.10200>
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- Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks <https://arxiv.org/pdf/2005.11464>
- Model Context Protocol <https://modelcontextprotocol.io/docs/getting-started/intro>
- Agentic Skills <https://geminicli.com/docs/cli/skills/> <https://agentskills.io/home>